



CIRCLES & CONICS

DAY-02 (16-03-24)

SECTION – I

(MULTIPLE CORRECT CHOICE TYPE)

1. A set of parabola(s) has their vertex at $(-9, 3)$ and it touches the x -axis at origin then the equation of axis of such parabola(s) can be
A) $x - y + 12 = 0$ B) $x - 2y + 15 = 0$
C) $2x - y + 3 = 0$ D) $x + y + 12 = 0$
2. Three circles of same radius r are drawn so that each one touches the other two externally. The radius of the circle which touches all the three circles is
A) $(3 + 2\sqrt{3})\frac{r}{3}$ B) $(2 - \sqrt{2})\frac{r}{2}$
C) $(2 - \sqrt{3})\frac{r}{\sqrt{3}}$ D) $(2 + \sqrt{2})\frac{r}{3}$
3. C_1, C_2 are two circles of radii $a, b (a < b)$ respectively touching both the coordinate axes and have their centres in the first quadrant. Then the true statements among the following are
A) If C_1, C_2 touch each other then $\frac{b}{a} = 3 + 2\sqrt{2}$
B) If C_1, C_2 are orthogonal then $\frac{b}{a} = 2 + \sqrt{3}$
C) If C_1, C_2 intersect in such a way that their common chord has maximum length then $\frac{b}{a} = 3$
D) If C_2 passes through centre of C_1 then $\frac{b}{a} = 2 + \sqrt{2}$

4. If the tangent at the point $\left(4\cos\theta, \frac{16}{\sqrt{11}}\sin\theta\right)$ to the ellipse $16x^2 + 11y^2 = 256$ is also a tangent to the circle $x^2 + y^2 - 2x = 15$, then θ is
- A) $\pi/3$ B) $2\pi/3$ C) $\pi/6$ D) $5\pi/3$
5. If $al^2 - bm^2 + 2dl + 1 = 0$ where a, b, d are fixed positive real numbers such that $a + b = d^2$, then the line $lx + my + 1 = 0$ touches a fixed circle. Then the fixed circle
- A) which cuts the x-axis orthogonally
B) with radius equal to b
C) On which the length of the tangent from the origin is $\sqrt{d^2 - b}$
D) with centre $(d, 0)$
6. The curves $x^2 + y^2 + 6x - 24y + 72 = 0$ & $x^2 - y^2 + 6x + 16y - 56 = 0$. Intersects in 4 points P, Q, R, S lying on parabola. Let A be focus of parabola. Then
- A) $AP + AQ + AR + AS = 20$ B) $AP + AQ + AR + AS = 40$
C) vertex $= (-3, -1/4)$ D) focus $= (-3, 1/4)$
7. If the normal at any given point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets its auxiliary circle at Q and R such that $\angle QOR = 90^\circ$, where O is the centre of ellipse, then
- A) $a^4 + 2b^4 \geq 3a^2b^2$ B) $a^4 + 2b^4 \geq 5a^2b^2 + 2a^3b$
C) $a^4 + 2b^4 \geq 3a^2b^2 + ab$ D) none of these
8. $(2, 6)$ and $(12, k)$ are the foci of an ellipse which touches both the coordinate axes. Then its
- A) Point of contact with x – axis is $(8, 0)$
B) Point of contact with y – axis is $\left(0, \frac{40}{7}\right)$

C) Eccentricity of the ellipse is $\frac{\sqrt{13}}{5}$

D) Length of latus rectum of the ellipse is $\frac{24\sqrt{2}}{5}$

9. A right angled triangle ABC, right angled at A is inscribed in hyperbola $xy = c^2$ ($c > 0$) such that slope of BC is 2. If distance of point A from centre of $xy = c^2$ is $\sqrt{10}$, then which of the following is/are correct for $xy = c^2$

A) the value of c is 2

B) the value of c is 4

C) the equation of normal at point A can be $y = 2x - 3\sqrt{2}$

D) the equation of normal at point A can be $y = 3x + 8\sqrt{2}$

10. Let E_1 and E_2 be two ellipses whose centers are at origin. The major axes of E_1 and E_2 lie along the x-axis and y-axis respectively. Let S be the circle $x^2 + (y-1)^2 = 2$. The straight line $x + y = 3$ touches the curves S, E_1 and E_2 at P, Q and R respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2 , respectively, then the correct expressions are

A) $e_1^2 + e_2^2 = \frac{43}{40}$

B) $e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$

C) $|e_1^2 - e_2^2| = \frac{5}{8}$

D) $e_1 e_2 = \frac{\sqrt{3}}{4}$

11. A circle C touches the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$, internally at points P and Q. The circle C has the center at $(1, 0)$. Then, which of the following is/are correct:-

A) radius of the circle C is $\sqrt{\frac{11}{3}}$ unit

B) length of the intercept made by the circle C on the Y-axis is $4\sqrt{\frac{2}{3}}$ unit

C) abscissa of points P & Q is $\frac{4}{3}$

D) ordinates of points P & Q are $\pm \frac{4\sqrt{2}}{3}$

12. Consider an ellipse with foci at (5,15) and (21,15). If the x-axis touches the ellipse, then which of the following is/are correct:-

A) the length of the minor axis of ellipse is 15 unit

B) the length of the major axis of ellipse is 34 unit

C) the length of the latus rectum of ellipse is $\frac{450}{17}$ unit

D) the eccentricity of the ellipse is $\frac{8}{17}$

13. From the point (1,1) a pair of tangents are drawn to the hyperbola $\frac{x^2}{4} - y^2 = 1$. Then, point(s) of contact lies in

A) 1st quadrant

B) 2nd quadrant

C) 3rd quadrant

D) 4th quadrant

14. The locus of a point $P(x, y)$ satisfy $\left| \sqrt{(x-1)^2 + (y-1)^2} - \sqrt{(x-2)^2 + (y-3)^2} \right| = K$; K being a constant, is

A) a line segment if $K = \sqrt{5}$

B) two rays if $K = \sqrt{5}$

C) a hyperbolic curve if $K \in (0, \sqrt{5})$

D) a line if $K = 0$

15. Points P and Q are taken on the ellipse $(x + y - 1)^2 + 2(x - y + 2)^2 = 8$. If $\alpha, \beta, \gamma, \delta$ are the lengths of the sides of quadrilateral PAQB where A and B are foci of the ellipse, then $\alpha^3 + \beta^3 + \gamma^3 + \delta^3$ is always greater than
- A) 14 B) 18 C) 28 D) 30
16. The value of 'a' for which two curves $y = ax^2 + ax + \frac{1}{24}$ and $x = ay^2 + ay + \frac{1}{24}$ touch each other is
- A) $\frac{2}{3}$ B) $\frac{3}{2}$ C) $-\frac{2}{3}$ D) $-\frac{3}{2}$
17. Circles of radii 3 and 4 intersect orthogonally the area common to both the circles is
- A) $8\pi + 12 - 7 \tan^{-1} \frac{4}{3}$ B) $8\pi - 12 - 7 \tan^{-1} \frac{4}{3}$
- C) $8\pi - 12 + 7 \tan^{-1} \frac{3}{4}$ D) $\frac{9\pi}{2} - 12 + 7 \tan^{-1} \frac{3}{4}$
18. Which of the following statements are true?
- A) The equation $x^2 \cos^2 \theta + y^2 \cot^2 \theta = 1$ represents a family of confocal ellipses
- B) The equation $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$ represents a family of confocal hyperbolas
- C) The shortest distance of a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to an asymptote of the hyperbola is 'b'
- D) The shortest distance of a focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to an asymptote of the hyperbola is 'a'
19. If the line $\sqrt{3}y - x = 3$ and $x - 2 = 0$ are tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then

A) Eccentricity of the ellipse is $\frac{1}{2}\sqrt{\frac{7}{3}}$

B) Eccentricity of the ellipse is $\sqrt{\frac{3}{7}}$

C) Length of Latus rectum of the ellipse is $\frac{5}{3}$

D) Area of the ellipse is $2\pi\sqrt{\frac{5}{3}}$ sq. units

20 Let A = (-1, 0) and B = (2, 0) be two points on the x – axis. A point 'M' (lying on only one side of AB) is moving in the xy – plane in such a way that $\angle MBA = 2\angle MAB$. Then the point 'M' moves along a conic whose

A) Eccentricity is 2

B) Eccentricity is $\frac{1}{2}$

C) Latus rectum is of length 6

D) Directrices are $x = \pm\left(\frac{1}{2}\right)$

SECTION –II

(INTEGER ANSWER TYPE)

1. The area of triangle formed by the chord of contact of P(3,2) to the circle

$x^2 + y^2 + 4x + 2y - 4 = 0$, With the pair of tangents from P is **D then** $\left(\frac{34D}{15}\right)^{\frac{1}{2}} =$

2. The difference between the radii of the largest and the smallest circles which have their centre on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$ and pass through the point (a,b) lying outside the given circle is .

3. Maximum number of common normals of $y^2 = 4ax$ and $x^2 = 4by$ is equal to

4. The length of the chord of the parabola $y^2 = x$ which is bisected at the point (2,1) is $a\sqrt{b}$ then $a+b =$
5. If $\sqrt{\lambda}$ is minimum distance between the curves $y^2 = 4x$ and $x^2 + y^2 - 12x + 31 = 0$ then λ
6. Tangents at three points of a parabola $y^2 = 4ax$ form an equilateral triangle. Then the locus of vertices of the triangle is $(K_1x + a)(K_2a + x) = K_3y^2$ then $K_1 + K_2 - K_3$
7. A line L: $y = mx + 3$ meets y-axis at P(0,3) and arc of the parabola $y^2 = 16x, 0 \leq y \leq 6$ at the point $Q(x_0, y_0)$. The tangent to the parabola at $Q(x_0, y_0)$ intersects the y-axis at $R(0, y_1)$. The slope m of the line L is chosen such that area of the triangle PQR has local maximum then $m + y_0 + y_1 + 2\Delta =$
[where Δ denotes maximum area of triangle PQR]
8. A parabola which have directrix $x + y + 2 = 0$ and touches a line $2x + y - 5 = 0$ at (2,1). If m is the latus rectum of the parabola, then the value of $\frac{m}{\sqrt{2}}$ is _____
9. Consider a triangle ABC, CC_1 and BB_1 are the medians drawn through the points B and C respectively and G is the centroid of the $\triangle ABC$. If the points A, C_1 , G and B_1 are concyclic, then value of $\frac{b^2 + c^2}{a^2}$ is ? ($a = BC, b = AC, c = AB$).
10. 27 are the length of any two perpendicular chords of a parabola, passing through the vertex of the parabola. If l is the length of latus rectum of the parabola, then write the value of $\frac{5l}{16}$.
11. $P_1, P_2, \dots, P_i, \dots, P_n$ are the points on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and $Q_1, Q_2, \dots, Q_i, \dots, Q_n$ are the corresponding points on the auxiliary circle of the ellipse. If the line joining

centre C to Q_i meets the normal at P_i w.r.t. the given ellipse at K_i and $\sum_{i=1}^n (CK_i) = 175$, then the value of $n/5$ is

12. Coordinate of the vertices B and C of triangle ABC are respectively (2, 0) and (8, 0). The vertex 'A' is varying in such a way that $\tan\left(\frac{B}{2}\right)\tan\left(\frac{C}{2}\right) = \frac{1}{4}$. Then the locus of 'A' is an ellipse whose semi-major axis is of length.

13. Let E_1 and E_2 be two ellipses. The area of the ellipse E_2 is one-third the area of quadrilateral formed by the tangents at the ends of latusrectum of the ellipse $E_3 (E_3 : 5x^2 + 9y^2 = 45)$. The eccentricities of E_1, E_2 and E_3 are equal. E_1 is inscribed in E_2 in such a way that both E_1 and E_2 touch each other at one end of their common major axis. If the length of major axis of E_1 is equal to the length of minor axis of E_2 , then area of ellipse E_2 outside the ellipse E_1 in square units is ____

14. Point "O" is the centre of an ellipse with major axis AB and minor axis CD. Point F is one focus of the ellipse. If $OF = 6$ and the diameter of the inscribed circle of the triangle OCF is 2 and the product of $(AB)(CD)$ is k. Then the value of $[\sqrt{k}] = \text{_____}$. Where $[.]$ denotes greatest integer function

15. The line $2x + y = 1$ is the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is:

16. Suppose that the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let P_1 and P_2 be the parabolas with a common vertex at $(0, 0)$ and with foci at $(f_1, 0)$ and $(2f_2, 0)$ respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and

T_2 be a tangent which passes through $(f_1, 0)$. If M_1 is the slope of T_1 and M_2 is the slope of T_2 , then the value of $\frac{1}{M_1^2} + M_2^2$ is

17. A circle has the same centre as an ellipse and passes through foci F_1 and F_2 of the ellipse, such that the two curves intersect at 4 points. Let P be any one of their point of intersection. If the major axis of the ellipse is 14 and the area of the triangle PF_1F_2 is 24, then the distance between the foci is K, then $K-8$ equals to
18. If a normal to a rectangular hyperbola at P meets the transverse axis at Q and foci of the hyperbola are S and S^1 , then $6\left(\frac{SP}{SQ}\right)^2$ is equal to
19. If a line L is a common tangent to the ellipse $16x^2 + 21y^2 = 336$ and the circle $x^2 + y^2 = r^2$ ($3 < r < \sqrt{21}$) and PQ is a chord of the circle parallel to L and passing a focus of the ellipse, then $PQ =$ _____
20. A chord of the rectangular hyperbola $xy = 16$ intersects the hyperbola at P, Q and x – axis, y- axis at R and S respectively and 'O' be centre of the hyperbola. If the area of ΔOPS is S_1 and the area of ΔQOR is S_2 , then the value of $\frac{4S_1S_2}{S_1^2 + S_2^2}$ is equal to

SOL. 4

$$\text{Ana of } \Delta OPS = \text{Ana of } \Delta OQR$$

21. If the two circles whose radius are 4, 9 are intersect at P and Q and a common tangent to both the circles touches the circles at A and B, then the circumradius of the triangle PAB is _____
22. A circle has the same centre as an ellipse, and passes through the foci F_1 and F_2 of the ellipse, such that the two curves intersect at 4 points. Let P be any one of their point of

intersection. If length of the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30 sq units and the distance between the foci F_1 & F_2 is k. then the value of k-4 is _____

SECTION-III

(Paragraph Type Questions)

Passage-1:-

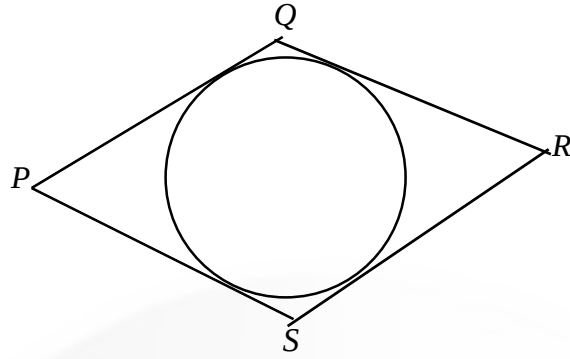
Consider a parabola $y^2 = 28x$. AB, CD are two perpendicular chords through focus of the parabola.

1. If AB makes angle $\alpha = \frac{\pi}{3}$ with x-axis, then $AB+CD =$
A) $\frac{48}{\sqrt{3}}$ B) $\frac{448}{3}$ C) $\frac{28}{3}$ D) $\frac{256}{3}$
2. If a quadrilateral ACBD is formed, then its minimum area = sq.units,
A) 84 B) $\frac{1568}{3}$ C) $\frac{448}{3}$ D) 1568
3. If length CD is $\frac{112}{3}$ then slope of chord AB is
A) $\frac{1}{\sqrt{3}}$ B) 1 C) $\sqrt{3}$ D) $2 - \sqrt{3}$

Passage-2:-

In a variable parallelogram PQRS radius of circle 'C' touching all its sides internally is $r = 256$, Now circles of integer radius are drawn on right side of C, Such that every circle touches externally the circle present to its left and also both the sides RQ, RS of the parallelogram and now same is repeated on left side of C.

i.e. Every circle on left side of C touches the circle to its right and also the sides PQ, PS



4. If in (n, r) , n = number of circles that can be drawn to the right of 'C' and r is the least radius ($r > 1$) among them, Find sum of all possible values of n
- A) 14 B) 7 C) 16 D) 8
5. As defined in the above question if (n_1, r_1) and (n_2, r_2) denote the right side and left side of C respectively, given $\frac{1}{r_1} + \frac{1}{r_2} = \frac{k}{r}$ and $n_1 + n_2 = l$ which of the following is not true ?
- A) For some value of k , two distinct values of l are possible
- B) For some value of l , two distinct values of k are possible
- C) l is always even
- D) $\frac{k}{l}$ is always an integer
6. If C_1, C_2 represent circles touching C on either side and the sides PQ, RS respectively and C_3, C_4 (with radii r_3, r_4) represent circles touching the pairs (C, C_1) and (C, C_2) and also their common tangents respectively then a possible rational value of $\sqrt{\frac{7\sqrt{r_3} - 4\sqrt{r_4}}{\sqrt{r_3} \cdot \sqrt{r_4}}} =$
- A) $\frac{3}{4}$ B) $\frac{9}{16}$ C) $\frac{4}{3}$ D) $\frac{16}{9}$

Passage-3:-

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6,5)$ to touch the parabola at Q and R . C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R . Both the circles C_1 and C_2 pass through the focus of the parabola.

7. Area of the ΔPQR equals

- A) $1/2$ B) 1 C) 2 D) $1/4$

8. Radius of the circle C_2 is

- A) $5\sqrt{5}$ B) $5\sqrt{10}$ C) $10\sqrt{2}$ D) $\sqrt{210}$

Passage-4:-

A normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at its point P , meets the Y -axis in Q .

9. If normal at P passes through a point $(2a, 0)$, then the range of the values of eccentricity of the hyperbola is

- A) $[\sqrt{2}, \infty)$ B) $(1, \sqrt{2}]$ C) $(1, \infty)$ D) none

10. As P varies on the hyperbola (with fixed a & b), then locus of the mid-point of PQ is a/an

- A) parabola B) circle C) ellipse D) hyperbola

Passage-5:-

A variable line is drawn through the point $P(-1,1)$ to meet the curve $xy = 1$ in the points A & B , Which lie on same side of point P . Let the locus of the mid-point of AB , lies on the curve $C = 0$.

11. The locus of curve $C = 0$ satisfies

- A) $(2x+1)(2y-1) = -2$ B) $(2x+1)(2y-1) = 1$
C) $(2x+1)(2y-1) = -1$ D) $(2x-1)(2y+1) = 1$

- ### Passage-6:-

13. Locus of circumcenter of triangle OAB is

14. Shortest distance between the line and hyperbola is

- ### Passage-7:-

15. All points P are always the points common to:

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16. The locus of P is symmetric about:

- A) origin B) the line $y = x$ C) $y - \text{axis}$ D) $x - \text{axis}$

17. The area of region of the point P is:

- A) $\sqrt{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right)$ B) $\sqrt{3} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right)$ C) $2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right)$ D) $3 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right)$

SECTION-IV

(Matrix Type Questions)

1. Let C_1 and C_2 be two circles whose equations are $x^2 + y^2 - 2x = 0$ and $x^2 + y^2 + 2x = 0$

$P(\lambda, \lambda)$ is a variable point. When $\lambda \in$

Column – I

Column – II

A) P lies inside C_1 but outside C_2

P) $(-\infty, -1) \cup (0, \infty)$

B) P lies inside C_2 but outside C_1

Q) $(-\infty, -1) \cup (1, \infty)$

C) P lies outside C_1 and outside C_2

R) $(-1, 0)$

D) P does not lie inside C_2

S) $(0, 1)$

2. Match Column-I with Column-II

Column I

Column II

A) The circum circle of ΔPQR , where P, Q, R are conormal

P) $2\sqrt{2}$

points on $y^2 = 12x$ always pass through (α, β) then $\alpha + \beta$

B) If the lines parallel to $y=2$ meet the parabola $y^2 = 12(x+3)$ and get reflected and pass through

Q) 2

The point (α, β) then $\alpha + \beta$

C) The length of latus rectum of parabola

$$x = t^2 + t + 1, y = t^2 - t + 1, t \in R$$

R) $\sqrt{2}$

D) If (α, β) is focus of $x^2 + y^2 + 2xy - 6x - 2y + 3 = 0$ $\sqrt{|\alpha|} + \sqrt{|\beta|}$

S) 0

3.

Column I	Column II
i) Variable tangents to $\frac{x^2}{4} - y^2 = 1$ cuts $\frac{x^2}{4} + y^2 = 1$ in points P & Q . Locus of the mid-point of PQ is $(x^2 + 4y^2)^2 = K(x^2 - 4y^2)$; then $K =$ _____	P) 1
ii) From a point $(0, 2)$; a pair of tangents are drawn to $x^2 - y^2 = 1$, whose magnitude of the product of slopes is _____	Q) 4
iii) A circle of radius 2 unit is drawn through the points in which $\frac{x^2}{3} - \frac{y^2}{2} = 1$ & $\frac{x^2}{a^2} + y^2 = 1$ intersect. Then, $a^2 =$ _____	R) 5
iv) If a variable line has intercepts on the X & Y axes as e and e' respectively ; where e & e' are the eccentricities of a hyperbola and its conjugate hyperbola, then this line always touches the circle $x^2 + y^2 = r^2$, where $r =$ _____	S) 6

Code:

	i)	ii)	iii)	iv)
A)	Q	R	P	S

B)	Q	R	S	P
C)	R	Q	P	S
D)	R	Q	S	P

4.

Column I	Column II
i) If vertices of a rectangle of maximum area inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, are extremities of latus rectum. Then, the eccentricity of the ellipse is _____	P) $\frac{2}{\sqrt{5}}$
ii) If extremities of diameter of the circle $x^2 + y^2 = 16$ are foci of an ellipse, the eccentricity of the ellipse, if its size is just sufficient to contain the circle, is _____	Q) $\frac{1}{\sqrt{2}}$
iii) If normal at point $(6, 2)$ to the ellipse passes through its nearest focus $(5, 2)$, having center at $(4, 2)$, then its eccentricity is _____	R) $\frac{1}{3}$
iv) If extremities of latus rectum of the parabola $y^2 = 24x$ are foci of an ellipse and if ellipse passes through the vertex of the parabola, then its eccentricity is _____	S) $\frac{1}{2}$

Code:

	i)	ii)	iii)	iv)
A)	Q	Q	S	P
B)	Q	R	P	S
C)	Q	Q	P	S
D)	R	Q	S	P

5. If the normals at $(x_i, y_i); i = 1, 2, 3, 4$; to the hyperbola $xy = 2$, pass through the point $(3, 4)$, then match the columns:

Column I	Column II
i) $x_1 + x_2 + x_3 + x_4 =$	P) - 4
ii) $y_1 + y_2 + y_3 + y_4 =$	Q) - 1
iii) $x_1 x_2 x_3 x_4 =$	R) 3
iv) $y_1 y_2 y_3 y_4 =$	S) 4

Code:

	i)	ii)	iii)	iv)
A)	S	R	P	P
B)	R	S	P	Q
C)	S	R	P	Q
D)	R	S	P	P

6. The equation of the ellipse with area $\frac{3\pi}{2}$, passing through the point of intersection of $x^2 + 4y^2 = 4$ and $x - 1 = 0$, having x-axis as its major axis and eccentricity equal to $\frac{\sqrt{3}}{2}$ is $ax^2 + 2hxy + by^2 + 2gx + 2fy - 2 = 0$ then match the following Column-I with Column-II

Column-I		Column-II	
A)	A	P)	0
B)	B	Q)	1
C)	$2h + f$	R)	-1
D)	G	S)	4

KEY

(MULTIPLE CORRECT CHOICE TYPE)

Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY
1.	AB	6.	BC	11.	ABCD	16.	AD
2.	AC	7.	AB	12.	BCD	17.	BD
3.	ABCD	8.	ABCD	13.	CD	18.	ABC
4.	AD	9.	AC	14.	BCD	19.	ACD
5.	ACD	10.	AB	15.	ABCD	20.	ACDs

(INTEGER ANSWER TYPE)

Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY
1.	5	6.	5	11.	5	16.	4	21.	6
2.	6	7.	8	12.	5	17.	2	22.	9
3.	1	8.	9	13.	4	18.	3		
4.	7	9.	6	14.	8	19.	8		
5.	3	10.	9	15.	2	20.	4		

(Paragraph Type Questions)

Q.No.	KEY	Q.No.	KEY	Q.No.	KEY	Q.No.	KEY
1.	B	6.	A	11.	C	16.	D
2.	D	7.	A	12.	C	17.	B
3.	A	8.	B	13.	C	18.	
4.	C	9.	B	14.	B	19.	
5.	D	10.	D	15.	A	20.	

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(Matrix Type Questions)

Q.No.	KEY	Q.No.	KEY
1.	A-S B-R C-Q D-P	6.	A-Q, B-S, C-P, D-R.
2.		7.	
3.	B	8.	
4.	A	9.	
5.	D	10.	